

Assignment–3

TRANSMISSION LINE PERFORMANCE CALCULATOR

Course: EE315 – Power Systems

Name: Siri Chandana Kondaparthi

Group No.: 11

Member No.: 04

1. AIM

To develop a generalized MATLAB program to calculate the sending-end voltage, current, power, power factor, voltage regulation, and transmission efficiency of a transmission line.

2. PROBLEM STATEMENT

A 50 Hz transmission line of length LL km has a series impedance of ZZ ohm/km and a shunt admittance of YY mho/km. The receiving-end load is given in MW at a specified receiving-end voltage and lagging power factor.

The objective is to classify the transmission line as short, medium, or long based on its length and calculate the corresponding transmission-line performance parameters using the appropriate model.

3. USED DATA

The individual values are obtained from the assignment definitions:

$$K = 4 - \text{Member No.} = 4 - 4 = 0$$

$$J = \text{last digit of Group No.} = 1$$

$$XX = \text{Group No.} = 11$$

$$m = |10 - \text{Member No.}| = |10 - 4| = 6$$

Therefore,

$$KJ0 = 010 \text{ km} = 10 \text{ km}$$

and

$$5 \times XX = 5 \times 11 = 55 \text{ MW}$$

$$0.m = 0.6$$

Table 1: Personalized Assignment Data

Parameter	Value
Group Number	11
Member Number	04
K	0
J	1
XX	11
m	6
Frequency, f	50 Hz
Line Length, L	10 km
Resistance, R	0.1 Ω /km
Reactance, X	1 Ω /km
Shunt Admittance, Y	2.8×10^{-6} mho/km
Receiving-end Load, P_R	55 MW
Receiving-end Voltage, V_R	220 kV
Receiving-end Power Factor	0.6 lagging

4. THEORY AND FORMULAS

4.1 Transmission Line Classification

The transmission line is classified according to its length as follows:

- Short line: $L < 80$ km
- Medium line: $80 \leq L < 250$ km
- Long line: $L \geq 250$ km

For the given line,

$$L = 10 \text{ km} < 80 \text{ km}$$

Therefore, the line is classified as a **Short Transmission Line**.

For a short transmission line, the shunt capacitance is neglected and the short-line model is used.

4.2 Series Impedance

The total series impedance is

$$Z = (R + jX)L$$

4.3 ABCD Parameters for Short Transmission Line

For the short-line model,

$$A = D = 1$$

$$B = Z$$

$$C = 0$$

4.4 Receiving-End Current

For a three-phase system,

$$I_R = \frac{P_R}{\sqrt{3}V_R \cos \phi_R}$$

where ϕ_R is the receiving-end power-factor angle.

$$\phi_R = \cos^{-1}(pf_R)$$

Since the receiving-end power factor is lagging, I_R lags V_R by ϕ_R .

4.5 Sending-End Quantities

The ABCD equations are

$$V_S = AV_R + BI_R$$

$$I_S = CV_R + DI_R$$

The sending-end complex power is

$$S_S = 3V_S I_S^*$$

Thus,

$$S_S = P_S + jQ_S$$

and the sending-end power factor is

$$pf_S = \frac{P_S}{|S_S|}$$

4.6 Voltage Regulation

The voltage regulation is

$$\%VR = \frac{V_{R,\text{no load}} - V_R}{V_R} \times 100$$

At no load,

$$V_{R,\text{no load}} = \left| \frac{V_S}{A} \right|$$

4.7 Transmission Efficiency

The transmission efficiency is

$$\eta = \frac{P_R}{P_S} \times 100$$

5. CALCULATIONS

5.1 Line Classification

$$L = 10 \text{ km}$$

Since

$$10 < 80$$

the line is a **Short Transmission Line**.

Hence, the short-line model is used.

5.2 Total Series Impedance

$$\begin{aligned} Z &= (0.1 + j1)(10) \\ &= \boxed{1 + j10 \Omega} \end{aligned}$$

The magnitude and angle are

$$|Z| = \sqrt{1^2 + 10^2} = 10.0499 \Omega$$

$$\angle Z = \tan^{-1} \left(\frac{10}{1} \right) = 84.289^\circ$$

Therefore,

$$\boxed{Z = 10.0499 \angle 84.289^\circ \Omega}$$

5.3 Total Shunt Admittance

The specified shunt admittance is

$$Y = j(2.8 \times 10^{-6})(10)$$

$$Y = j2.8 \times 10^{-5} \text{ S}$$

Since this is a short transmission line, shunt admittance is neglected in the short-line model.

5.4 Receiving-End Phase Voltage

The receiving-end line-to-line voltage is 220 kV.

Therefore,

$$V_R = \frac{220}{\sqrt{3}}$$

$$\boxed{V_R = 126.872 \text{ kV/phase}}$$

5.5 Receiving-End Power Factor Angle

$$\phi_R = \cos^{-1}(0.6)$$

$$\boxed{\phi_R = 53.130^\circ}$$

Since the power factor is lagging, the current angle is -53.130° .

5.6 Receiving-End Current

$$I_R = \frac{55 \times 10^6}{\sqrt{3}(220 \times 10^3)(0.6)}$$

$$\boxed{|I_R| = 240.563 \text{ A}}$$

Thus,

$$I_R = 240.563 \angle (-53.130^\circ) \text{ A}$$

or approximately,

$$I_R = 144.338 - j192.397 \text{ A}$$

5.7 ABCD Parameters

For a short transmission line,

$$A = D = 1$$

$$B = 1 + j10 \, \Omega = 10.0499 \angle 84.289^\circ \, \Omega$$

$$C = 0 \, \text{S}$$

The ABCD determinant check is

$$AD - BC = (1)(1) - (1 + j10)(0) = 1$$

Therefore,

$$AD - BC = 1$$

5.8 Sending-End Voltage

Using

$$V_S = AV_R + BI_R$$

and the receiving-end phase voltage and current,

$$V_S = V_R + ZI_R$$

The calculated sending-end line-to-line voltage is

$$|V_S| = 223.594 \, \text{kV}$$

5.9 Sending-End Current

For a short line,

$$I_S = CV_R + DI_R$$

Since $C = 0$ and $D = 1$,

$$I_S = I_R$$

Hence,

$$|I_S| = 240.563 \text{ A}$$

5.10 Sending-End Power

The sending-end complex power is

$$S_S = 3V_S I_S^*$$

The calculated values are

$$P_S = 55.174 \text{ MW}$$

and

$$Q_S = 75.069 \text{ MVAR}$$

The apparent power is approximately

$$|S_S| = 93.168 \text{ MVA}$$

Therefore, the sending-end power factor is

$$pf_S = \frac{55.174}{93.168}$$

$$pf_S = 0.5922 \text{ lagging}$$

5.11 Voltage Regulation

At no load,

$$V_{R,\text{no load}} = \left| \frac{V_S}{A} \right|$$

Since $A = 1$,

$$V_{R,\text{no load}} = |V_S|$$

Thus,

$$V_{R,\text{no load}} = 223.594 \text{ kV}$$

and

$$\%VR = \frac{223.594 - 220}{220} \times 100$$

$$\boxed{\%VR = 1.634\%}$$

5.12 Transmission Efficiency

$$\eta = \frac{55}{55.174} \times 100$$

Therefore,

$$\boxed{\eta = 99.685\%}$$

6. MATLAB IMPLEMENTATION

The MATLAB program is generalized and takes the required line parameters from the user at run time. It automatically classifies the line according to its length and selects the appropriate transmission-line model.

The inputs are entered in the same command-window style as the reference result.

Listing 1: Generalized MATLAB Program

```

1  % Assignment-3: Transmission Line Performance Calculator
2  % Course: EE315 - Power Systems
3  % Name: Siri Chandana Kondaparthi
4  % Group No.: 11
5  % Member No.: 4
6
7  clc;
8  clear;
9  format compact;
10
11 % ----- USER INPUTS -----
12 L = input('Enter length of line, L (km) : ');
13 R = input('Enter resistance, R (ohm/km) : ');
14 X = input('Enter reactance, X (ohm/km) : ');
15 Yc = input('Enter shunt admittance magnitude, Y (mho/km) : ');
16 PR = input('Enter receiving end load, PR (MW) : ');
17 VR_kV = input('Enter receiving end voltage, VR (kV, L-L) : ');
18 pfR = input('Enter receiving end power factor (lagging) : ');
19
20 % ----- INPUT VALIDATION -----
21 if L <= 0
22     error('Line length must be greater than zero.');
```

$$Z = (R + jX) * L;$$

```

23 end
24
25 if VR_kV <= 0
26     error('Receiving-end voltage must be greater than zero.');
```

$$Y = 1j * Yc * L;$$

```

27 end
28
29 if PR < 0
30     error('Receiving-end load cannot be negative.');
```

```

48
49 % ----- LINE CLASSIFICATION -----
50 if L < 80
51     lineType = 'SHORT LINE';
52
53     % Short transmission line model
54     A = 1;
55     B = Z;
56     C = 0;
57     D = 1;
58
59 elseif L < 250
60     lineType = 'MEDIUM LINE';
61
62     % Medium transmission line - nominal-pi model
63     A = 1 + Y*Z/2;
64     B = Z;
65     C = Y*(1 + Y*Z/4);
66     D = A;
67
68 else
69     lineType = 'LONG LINE';
70
71     % Long transmission line - exact distributed model
72     gamma = sqrt(Z*Y);
73     Zc = sqrt(Z/Y);
74
75     A = cosh(gamma);
76     B = Zc*sinh(gamma);
77     C = (1/Zc)*sinh(gamma);
78     D = A;
79 end
80
81 % ----- SENDING-END QUANTITIES -----
82 VS = A*VR + B*IR;
83 IS = C*VR + D*IR;
84
85 Ssend = 3*VS*conj(IS);
86
87 PS = real(Ssend)/1e6;
88 QS = imag(Ssend)/1e6;
89 SS = abs(Ssend)/1e6;
90 pfS = abs(PS)/SS;
91
92 if QS >= 0
93     pfS_text = 'lagging';
94 else

```

```

95     pfS_text = 'leading';
96 end
97
98 VS_LL_kV = abs(VS)*sqrt(3)/1e3;
99 IS_A = abs(IS);
100
101 % ----- VOLTAGE REGULATION -----
102 VR_noload = VS/A;
103 VR_noload_LL_kV = abs(VR_noload)*sqrt(3)/1e3;
104
105 VR_percent = ((VR_noload_LL_kV - VR_kV)/VR_kV)*100;
106
107 % ----- TRANSMISSION EFFICIENCY -----
108 efficiency = (PR/PS)*100;
109
110 % ----- ABCD CHECK -----
111 check = A*D - B*C;
112
113 % ----- DISPLAY -----
114 fprintf('\n');
115 fprintf('-----\n');
116 fprintf('Based on length = %.2f km, the line is classified as: %s\n', ...
117         L, lineType);
118 fprintf('-----\n');
119
120 fprintf('\n');
121 fprintf('ABCD Parameters of the line:\n');
122
123 fprintf('A = %.4f /_%.3f deg (per unit)\n', ...
124         abs(A), angle(A)*180/pi);
125
126 fprintf('B = %.4f /_%.3f deg (ohm)\n', ...
127         abs(B), angle(B)*180/pi);
128
129 fprintf('C = %.6f /_%.3f deg (mho)\n', ...
130         abs(C), angle(C)*180/pi);
131
132 fprintf('D = %.4f /_%.3f deg (per unit)\n', ...
133         abs(D), angle(D)*180/pi);
134
135 fprintf('\n');
136 fprintf('Check: AD - BC = %.4f + j%.4f (should be ~1)\n', ...
137         real(check), imag(check));
138
139 fprintf('\n');
140 fprintf('===== RESULTS =====\n');
141

```

```

142 fprintf('Sending end voltage Vs = %.3f kV (line)\n', VS_LL_kV);
143 fprintf('Sending end current Is = %.3f A\n', IS_A);
144 fprintf('Sending end power Ps = %.3f MW\n', PS);
145 fprintf('Sending end reactive Q_s = %.3f MVAR\n', QS);
146 fprintf('Sending end p.f. = %.4f (%s)\n', pfS, pfS_text);
147 fprintf('Voltage Regulation = %.3f %%\n', VR_percent);
148 fprintf('Transmission Efficiency = %.3f %%\n', efficiency);
149
150 fprintf('=====\n');

```

7. MATLAB RESULT

```

Command Window

Enter length of line, L (km)           : 10
Enter resistance, R (ohm/km)           : 0.1
Enter reactance, X (ohm/km)           : 1
Enter shunt admittance magnitude, Y (mho/km): 2.8e-6
Enter receiving end load, PR (MW)      : 55
Enter receiving end voltage, VR (kV, L-L) : 220
Enter receiving end power factor (lagging) : 0.6

-----
Based on length = 10.00 km, the line is classified as: SHORT LINE
-----

ABCD Parameters of the line:
A = 1.0000 /_0.000 deg (per unit)
B = 10.0499 /_84.289 deg (ohm)
C = 0.000000 /_0.000 deg (mho)
D = 1.0000 /_0.000 deg (per unit)

Check: AD - BC = 1.0000 + j0.0000 (should be ~1)

===== RESULTS =====
Sending end voltage Vs = 223.594 kV (line)
Sending end current Is = 240.563 A
Sending end power Ps = 55.174 MW
Sending end reactive Q_s = 75.069 MVAR
Sending end p.f. = 0.5922 (lagging)
Voltage Regulation = 1.634 %
Transmission Efficiency = 99.685 %
=====

fx >> |

```

Figure 1: Results from MATLAB

8. RESULT SUMMARY

Table 2: Final Calculated Results

Parameter	Result
Line Classification	Short Transmission Line
Model Used	Short-line model
Total Series Impedance	$1 + j10 \Omega$
A	1
B	$1 + j10 \Omega$
C	0 S
D	1
Sending-end Voltage, V_S	223.594 kV
Sending-end Current, I_S	240.563 A
Sending-end Active Power, P_S	55.174 MW
Sending-end Reactive Power, Q_S	75.069 MVAR
Sending-end Power Factor	0.5922 lagging
Voltage Regulation	1.634%
Transmission Efficiency	99.685%

9. CONCLUSION

The generalized MATLAB program successfully classifies the transmission line and calculates its sending-end performance parameters.

the calculated line length is 10 km. Since the line length is less than 80 km, it is classified as a **Short Transmission Line**, and the short-line model is used.

The calculated sending-end voltage is

$$\boxed{223.594 \text{ kV}}$$

the sending-end current is

$$\boxed{240.563 \text{ A}}$$

the sending-end power factor is

0.5922 lagging

the voltage regulation is

1.634%

and the transmission efficiency is

99.685%

Thus, the required transmission-line performance parameters were successfully calculated using the generalized MATLAB program.

10. GIT HUB

<https://github.com/K-SiriChandana/POWER-SYSTEMS-ASSIGNMENT3>